An LLM-Driven Pipeline for Synthetic Surveillance Video Dataset Generation in Virtual Environments

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Figure 1: Teaser. Example frames from a synthetic surveillance scenario generated by the proposed pipeline (approach–interact–leave) under a fixed CCTV-like viewpoint.

Abstract

Collecting real-world surveillance data for rare and security-critical behaviors is challenging due to ethical, legal, and practical constraints. In this work, we present a behavior-driven simulation pipeline for generating controllable surveillance scenarios in virtual environments. Built on Unreal Engine and MetaHuman actors, the proposed system combines graph-based spatial constraints with an LLM-driven semantic behavior planner to generate executable human behavior sequences.

Unlike recent text-to-video generation approaches that primarily focus on pixel-level realism, our framework emphasizes structured behavior execution, reproducibility, and explicit control over agent roles, routes, and interactions. Scenarios generated by the proposed framework are rendered from multiple fixed, CCTV-like viewpoints and paired with structured annotations.

The proposed framework enables the generation of synthetic surveillance data with structured annotations, including temporally segmented actions and object interactions, through role-conditioned human behavior simulation under explicit spatial and semantic constraints.

Index Terms: Virtual environments; behavior simulation; synthetic surveillance data; large language models; human action generation.

1 Introduction

Surveillance action recognition systems increasingly need to detect not only common activities but also rare and security-critical events such as covert exchanges, theft, or suspicious loitering. Acquiring real-world data for such behaviors remains difficult due to ethical concerns, privacy regulations, and the inherent rarity of abnormal events, which collectively hinder large-scale data collection and reliable annotation. Existing benchmarks such as Kinetics and AVA have advanced video understanding but primarily focus on everyday or sports-related activities, offering limited coverage of surveillance-specific behaviors [3, 2].

Synthetic data generation has therefore emerged as a practical alternative, enabling controllable environments and dense ground-truth annotations without manual labeling [5, 6]. Game engines such as Unreal Engine further support this direction by providing programmatic access to scene layouts, cameras, and virtual actors, enabling reproducible and scalable content generation [4, 1]. However, authoring executable human behavior scenarios in virtual environments remains challenging, as many systems rely on rule-based scripting or manually authored animations, limiting scalability and semantic diversity.

In this work, we introduce an LLM-driven behavior simulation pipeline for generating controllable surveillance scenarios in virtual environments. The proposed framework separates spatial feasibility from semantic intent by combining graph-based route construction with an LLM-driven semantic behavior planner, enabling executable and role-consistent human behaviors under explicit navigational and environmental constraints. This design supports VR-based behavior simulation with synthetic surveillance data generation as a particularly practical outcome.

2 System Overview

Figure 2 illustrates the overall pipeline. We construct surveillance-oriented indoor and outdoor environments in Unreal Engine and formalize navigable spaces as graphs consisting of spawn points, waypoints, endpoints, and interactable objects. Actors are instanti-

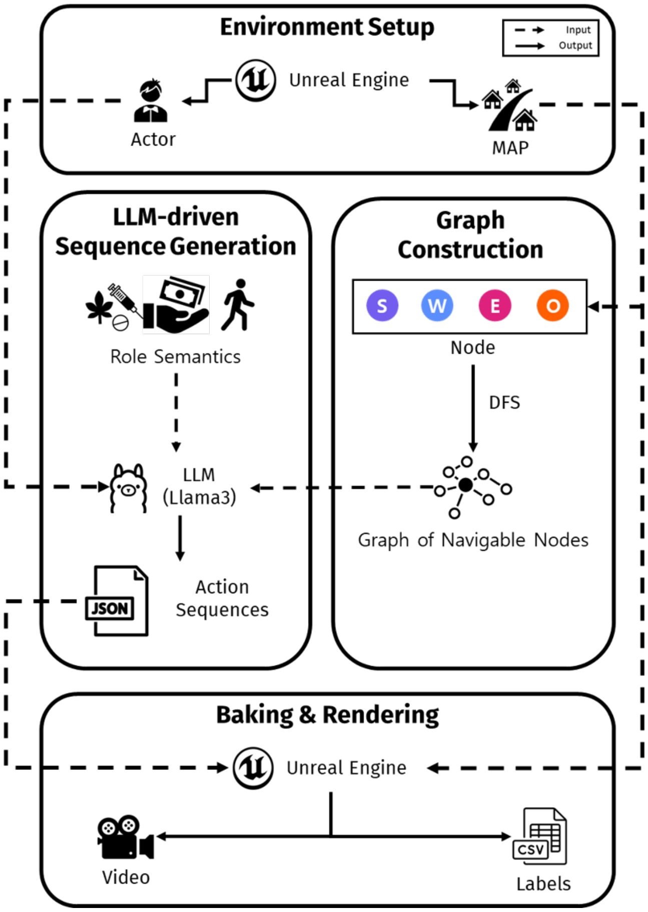

Figure 2: Overview of the proposed pipeline for LLM-driven surveillance scenario generation in virtual environments.

ated using MetaHuman models with appearance variations to support identity consistency across views.

Given an assigned role (e.g., buyer, dealer, bystander) and a graph-based route, an LLM-driven controller generates structured action sequences specifying movement styles, arrival behaviors, and object interactions. These sequences are automatically translated into Unreal Engine LevelSequences and rendered from multiple fixed camera viewpoints, mimicking CCTV layouts. During rendering, frame-level annotations (actor IDs, roles, bounding boxes, and action labels with temporal boundaries) are programmatically exported, eliminating manual labeling effort and ensuring consistent annotation quality.

In our framework, the environment graph and the LLM controller play complementary roles in agent behavior generation. The graph enforces physically feasible navigation under spatial constraints, while the LLM acts as a semantic behavior planner that determines how the agent behaves along the route.

3 LLM-Driven Scenario Generation

Unlike rule-based scripting systems that require handcrafted mappings between locations and actions, our LLM-based controller enables flexible yet constrained behavior generation. We combine schema-level constraints (e.g., required object interactions for specific roles) with role-dependent style biases to generate diverse but semantically consistent behaviors along the same route.

Before translation into Unreal Engine LevelSequences, generated action sequences are validated for animation compatibility, temporal consistency, and interaction validity, with corrective strategies applied when necessary to ensure physical plausibility.

4 Applications of Behavior-Driven VR Simulation

The proposed pipeline enables systematic instantiation of surveillance-oriented human behaviors under explicit control over agent roles, routes, and interactions. Using this framework, we generated 463 synthetic surveillance scenarios featuring abnormal events such as dead-drop exchanges, rendered from multiple fixed viewpoints with structured annotations supporting downstream surveillance tasks.

5 Limitations and Future Work

This extended abstract emphasizes the system and dataset generation capabilities rather than downstream model evaluation. Future work will (i) evaluate action recognition models trained on the generated synthetic data, (ii) compare LLM-driven generation against simpler scripted baselines, and (iii) expand scenario complexity via multi-actor interactions and broader environment diversity. Recent text-to-video world models can generate visually realistic videos from language prompts; however, they offer limited controllability, reproducibility, and support for explicit agent interactions. In contrast, our system prioritizes executable behavior simulation within virtual environments, enabling structured control over agent actions, interactions, and annotations rather than pixel-level realism.

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